

Joshua Garcia Nieves

Cidra, Puerto Rico • imjoshuagarcia1212@gmail.com
yoshimaster.com • github.com/Y0shm4st3r • linkedin.com/in/joshua-garcia-nieves
Full version with complete contact details available on request.

PROFILE

Second-year Computer Engineering student seeking a summer internship or co-op in IT infrastructure, networking or systems. I build and operate the self-hosted infrastructure my household uses every day (a segmented network, two hypervisors, three ZFS pools and about twenty services) and document every failure, including the ones I have not solved yet.

EDUCATION

B.S. Computer Engineering | Universidad Ana G. Méndez - Gurabo, PR

Aug 2025 - Present

- GPA: 3.89 / 4.00

TECHNICAL PROJECTS

Segmented Home Network (Homelab)

2024 - Present

- Designed a network segmented with 802.1Q VLANs (Servers, Home, Cameras) on a MikroTik managed switch, with a virtualized OPNsense firewall routing between them over a single trunk (router-on-a-stick).
- Wrote least-privilege inter-VLAN rules: cameras reach only the video server over RTSP and have no internet access.
- Routed a single host through a WireGuard VPN with policy-based routing, and verified the kill switch by taking the tunnel down and confirming no traffic leaked.
- Diagnosed a game console's internet outage as a routing problem, not a firewall one: the DNS server in its DHCP reservation matched the VPN monitor IP, so a /32 route sent its traffic into the tunnel. Ruled out six hypotheses first.
- Restricted switch management (SSH/Winbox) to a dedicated management port and published self-hosted services through Cloudflare Tunnel.

Virtualization and Storage Server

2024 - Present

- Administer two Proxmox VE hosts running VMs and LXC containers, with GPU passthrough (RTX 3060) for real-time object detection in Frigate.
- Run TrueNAS SCALE with three ZFS pools, including a 4-disk RAIDZ2, shared over NFS; track disk health with SMART by world-wide name and scheduled scrubs.
- Configured Network UPS Tools across three machines so that on battery, VMs shut down before their hosts to protect ZFS pool integrity.
- Deployed about twenty Docker services (Immich, Nextcloud, Jellyfin, Frigate, Ollama), monitored per host with Netdata and Uptime Kuma.

Troubleshooting Log: "Where It Broke"

2026

- Logged over 170 incidents across my network, homelab and Linux machines, each tagged with the layer where the symptom appeared and the layer of the root cause, including the ones still open. In 58 of them the cause sat in a different layer than the symptom.
- Published a sanitized version on my portfolio; the full record with evidence and discarded hypotheses lives in a private repository.

3D Printer Firmware Rebuild

2026

- Compiled Marlin 2.1.4 from source for an STM32 board with a BLTouch probe; rebuildable from two configuration files, with the flashed binary traced by SHA-256 and the build timestamp the machine reports.
- Separated a 1.938 mm bed error by least-squares plane fitting into correctable tilt and a 0.163 mm non-planar residual; re-leveling cut the range to 0.180 mm with no new parts.

TECHNICAL SKILLS

- **Networking:** 802.1Q VLANs & trunking, inter-VLAN firewalling (OPNsense), MikroTik RouterOS, WireGuard, policy-based routing, NAT, DHCP/DNS (Unbound), Cloudflare Tunnel
- **Systems:** Proxmox VE, LXC, Docker Compose, GPU passthrough, Linux (Debian, Arch), systemd, Git
- **Storage & Monitoring:** TrueNAS SCALE, ZFS (RAIDZ2, mirror), NFS, smartctl, NUT, Netdata, Uptime Kuma, MTR
- **Hardware & Firmware:** Marlin, PlatformIO, soldering, breadboarding, PC assembly, FDM 3D printing, Fusion 360
- **Also familiar:** Tailscale, Bash, Windows 10/11, Microsoft Office, Google Workspace; Python (learning)
- **Languages:** Spanish (native), English (intermediate)

EXPERIENCE

Overnight Stocker | Walmart Supercenter - Cayey, PR

Jun 2023 - Present

- Restock and organize merchandise overnight in a high-volume store and correct inventory discrepancies, coordinating with the team to meet opening deadlines.

Kitchen / Production Team | Burger King - Caguas, PR

Nov 2020 - Apr 2021

- Prepared orders to quality and time standards in a fast-paced kitchen, following food safety protocols.